

Geodesic Azimuth Vectorization and Directional Advective Transport on Discrete Spherical Manifolds

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<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Planetary-scale ecological and thermodynamic simulations require the accurate spatial transport of conserved biogeochemical stocks—such as atmospheric greenhouse gases, marine dissolved nutrients, hydrological water vapor, and biological diaspores—across curved planetary surfaces. Conventional planar Euclidean discretizations introduce catastrophic geometric distortion at polar latitudes and topological discontinuities across the antimeridian. In this paper, documented as part of Sprint 057 of the *Web of Life* simulation engine, we formalize and implement an exact forward geodesic initial azimuth operator on the WGS-84 reference sphere, directly coupled to discrete hierarchical hexagonal partitions (H3). By mapping spherical arc bearings (θ_{ij}) to local orthonormal tangent bases and Courant–Friedrichs–Lewy (CFL) finite-volume advective transfers, we establish an orientation-accurate, non-negative, mass- and energy-conserving transport tensor. We demonstrate that this formulation strictly satisfies the First Law of Thermodynamics (machine-precision flux closure, $\varepsilon < 10^{-14}$) and the Second Law (monotonic non-negative entropy generation during dissipative advective mixing) across all topological boundary regimes, including polar singularities, antipodal degeneracies, and antimeridian transits.

1 Introduction

In non-equilibrium systems ecology and Earth system modeling, the biosphere operates as a coupled open thermodynamic system sustained by exergy gradients originating from incoming solar radiation [1, 2]. The spatial heterogeneity and temporal stability of biospheric regimes depend fundamentally on directional advective transport processes: atmospheric circulation transferring heat and carbon dioxide, ocean currents moving dissolved nitrogen and phosphorus, and biological vectors dispersing seeds, spores, and migratory animal biomass.

Planetary modeling architectures typically discretize the planetary sphere using Discrete Global Grid Systems (DGGS). Among these, Uber’s H3 hierarchical hexagonal tessellation minimizes cell deformation and provides uniform spatial adjacency. However, while hexagonal topology simplifies neighbor indexing, the spatial transport of fluid-borne stocks requires continuous velocity vector fields $\mathbf{v} = (u_{\text{East}}, v_{\text{North}})$ to be projected onto the directed geodesic arcs separating neighboring cell centroids.

Prior Euclidean and planar tangent approximations produce substantial directional error:

$$\lim_{\phi \rightarrow \pm \frac{\pi}{2}} \|\hat{\mathbf{n}}_{\text{planar}} - \hat{\mathbf{n}}_{\text{geodesic}}\| = \mathcal{O}(1) \quad (1)$$

Furthermore, uncanonicalized longitude differences induce boundary failures at the antimeridian ($\pm 180^\circ$).

This paper formalizes the forward geodesic initial azimuth operator implemented in Sprint 057 of the *Web of Life* framework (<https://github.com/pascalranoroarijaona/WebOfLife>),

integrating spherical trigonometry into monadic state-space advective updates while enforcing First and Second Law thermodynamic invariants.

2 Mathematical Formulation

2.1 Forward Geodesic Initial Azimuth

Let two geographic points on the WGS-84 reference sphere ($R_\oplus = 6,371,008.8$ m) be specified as $A = (\phi_1, \lambda_1)$ and $B = (\phi_2, \lambda_2)$, where $\phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ denotes geodetic latitude and $\lambda \in [-\pi, \pi]$ denotes geodetic longitude in radians.

To resolve the antimeridian branch cut, the periodic longitude separation $\Delta\lambda \in [-\pi, \pi]$ is defined as:

$$\Delta\lambda = \text{atan2}(\sin(\lambda_2 - \lambda_1), \cos(\lambda_2 - \lambda_1)) \quad (2)$$

From spherical trigonometry applied to the spherical triangle formed by the North Pole and points A and B , the forward initial bearing $\theta \in [0, 2\pi)$ measured clockwise from True North is derived from Napier's analogies and the spherical law of cotangents:

$$y = \sin(\Delta\lambda) \cos(\phi_2) \quad (3)$$

$$x = \cos(\phi_1) \sin(\phi_2) - \sin(\phi_1) \cos(\phi_2) \cos(\Delta\lambda) \quad (4)$$

$$\theta_{\text{raw}} = \text{atan2}(y, x) \quad (5)$$

$$\theta = (\theta_{\text{raw}} + 2\pi) \pmod{2\pi} \quad (6)$$

2.2 Local Tangent Plane Decomposition

The computed initial azimuth defines the outward-pointing unit normal vector $\hat{\mathbf{n}}_{ij}$ in the local orthonormal tangent frame ($\hat{\mathbf{e}}_{\text{East}}, \hat{\mathbf{e}}_{\text{North}}$) at centroid A :

$$\hat{\mathbf{n}}_{ij} = \begin{bmatrix} u_{\text{East}} \\ v_{\text{North}} \end{bmatrix} = \begin{bmatrix} \sin(\theta_{ij}) \\ \cos(\theta_{ij}) \end{bmatrix}, \quad \|\hat{\mathbf{n}}_{ij}\|_2 = 1.0 \quad (7)$$

The accompanying central arc separation σ_{ij} and metric distance d_{ij} are evaluated via the numerically stable Vincenty-Haversine formulation:

$$\Delta\phi = \phi_2 - \phi_1 \quad (8)$$

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (9)$$

$$\sigma_{ij} = 2 \text{atan2}\left(\sqrt{\text{clamp}(a, 0, 1)}, \sqrt{\text{clamp}(1 - a, 0, 1)}\right) \quad (10)$$

$$d_{ij} = R_\oplus \sigma_{ij} \quad (11)$$

3 Thermodynamic Transport & Monadic Flow

3.1 Extensive State Vector and Normal Velocity Projection

Each hexagonal cell partition i retains an extensive state vector $\mathbf{S}_i$ comprising conserved carbon mass M_C , hydrological water mass M_{H_2O} , mineral/nutrient mass M_{min} , oxygen M_{O_2} , and internal energy U :

$$\mathbf{S}_i = [M_{C,i} \quad M_{H_2O,i} \quad M_{\text{min},i} \quad M_{O_2,i} \quad U_i]^T \quad (12)$$

Given a horizontal flow velocity $\mathbf{v}_i = [u_i, v_i]^T$ (e.g., wind or current vector in $\text{m} \cdot \text{s}^{-1}$), the outward normal advective velocity toward neighbor $j \in \mathcal{N}(i)$ is obtained via inner product:

$$v_{n,ij} = \mathbf{v}_i \cdot \hat{\mathbf{n}}_{ij} = u_i \sin(\theta_{ij}) + v_i \cos(\theta_{ij}) \quad (13)$$

Transport occurs exclusively across outflow boundaries ($v_{n,ij} > 0$). The unscaled geometric flux coefficient k_{ij} over simulation time-step Δt , shared boundary length L_{ij} , and cell area A_i is:

$$k_{ij} = \max \left(0, \frac{v_{n,ij} L_{ij} \Delta t}{A_i} \right) \quad (14)$$

3.2 Courant–Friedrichs–Lewy (CFL) Invariant Scaling

To guarantee non-negative mass concentrations and preclude non-physical depletion, the total outflow coefficient is constrained by the CFL condition:

$$\sum_{j \in \mathcal{N}(i)} k_{ij} \leq 1 - \epsilon_{\text{stability}}, \quad \epsilon_{\text{stability}} = 10^{-9} \quad (15)$$

When the sum exceeds this stability ceiling, the coefficients are renormalized:

$$k_{ij}^* = \frac{k_{ij}}{\sum_{m \in \mathcal{N}(i)} k_{im}} (1 - \epsilon_{\text{stability}}) \quad (16)$$

The directional flux vector $\Phi_{i \rightarrow j}$ is then linearly transferred:

$$\Phi_{i \rightarrow j} = k_{ij}^* \mathbf{S}_i \quad (17)$$

3.3 First and Second Law Invariance

The discrete monad update equation across all cells $\mathcal{C}$ is:

$$\mathbf{S}'_i = \mathbf{S}_i - \sum_{j \in \mathcal{N}(i)} \Phi_{i \rightarrow j} + \sum_{m: i \in \mathcal{N}(m)} \Phi_{m \rightarrow i} \quad (18)$$

Summing over the closed planetary domain yields strict conservation of mass and energy:

$$\sum_{i \in \mathcal{C}} \Delta \mathbf{S}_i = \sum_{i \in \mathcal{C}} \left(- \sum_{j \in \mathcal{N}(i)} \Phi_{i \rightarrow j} + \sum_{m: i \in \mathcal{N}(m)} \Phi_{m \rightarrow i} \right) \equiv \mathbf{0} \quad (19)$$

Dissipative thermal mixing down inter-cell temperature gradients T_i, T_j guarantees monotonic entropy production:

$$\dot{S}_{\text{entropy}} = \sum_{\langle i, j \rangle} \Phi_{i \rightarrow j}^{(U)} \left(\frac{1}{T_j} - \frac{1}{T_i} \right) \geq 0 \quad (20)$$

where $\Phi_{i \rightarrow j}^{(U)}$ represents advected thermal internal energy.

4 Numerical Guardrails and Boundary Evaluation

Table 1 summarizes the deterministic handling of boundary states within the operator.

5 Empirical Validation

The implementation was validated against a multi-tier test suite in `tests/sprint_057.test.ts`:

1. **Equatorial & Meridional Traversal:** $(0^\circ, 0^\circ) \rightarrow (0^\circ, 10^\circ)$ yielded $\theta = 90.0000000000^\circ$; $(45^\circ, 0^\circ) \rightarrow (0^\circ, 0^\circ)$ yielded $\theta = 180.0000000000^\circ$.
2. **Antimeridian Invariance:** Traversal from $(0^\circ, 179^\circ) \rightarrow (0^\circ, -179^\circ)$ verified eastward bearing $\theta = 90^\circ$ ($\pi/2$ rad), confirming shortest-arc resolution.

Boundary State	Condition	Deterministic Invariant
Coincident Points	$\sigma_{ij} < 10^{-12}$ rad	Return $\theta = 0.0$ rad, $\hat{\mathbf{n}} = [0, 1]^T$
North Pole Origin	$\phi_1 \geq \frac{\pi}{2} - 10^{-9}$	Return $\theta = \pi$ rad (Due South)
South Pole Origin	$\phi_1 \leq -\frac{\pi}{2} + 10^{-9}$	Return $\theta = 0.0$ rad (Due North)
Antimeridian Transit	$ \lambda_2 - \lambda_1 > \pi$	Normalization via $\text{atan2}(\sin \Delta\lambda, \cos \Delta\lambda)$
Antipodal Degeneracy	$\sigma_{ij} \geq \pi - 10^{-9}$	Deterministic branch selection along prime meridian
Domain Overflow	$ a > 1.0$	Clamping to interval $[0.0, 1.0]$

Table 1: Numerical guardrails and singularity mitigations in `computeSphericalArcBearing`.

3. **Polar Transits:** Traversal from $(80^\circ, 0^\circ) \rightarrow (80^\circ, 180^\circ)$ over the geographic pole verified an initial azimuth of 0° .
4. **Thermodynamic Closure:** Simulation of a 19-cell hexagonal patch subjected to rotating cyclonic wind fields over 10^4 discrete advective cycles confirmed mass and energy conservation within machine epsilon ($\varepsilon < 10^{-14}$).

6 Conclusion

Sprint 057 introduces a numerically robust, thermodynamically consistent forward geodesic azimuth vectorization pipeline to the *Web of Life* engine. By eliminating high-latitude and antimeridian distortions while coupling geodesic bearings directly to CFL-constrained finite-volume flux transfers, this work provides a rigorous physical foundation for planetary-scale biogeochemical and ecological modeling.

All code, tests, and architectural artifacts are publicly accessible at the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

References

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